
Data Science Final Project: Laptop Price Analysis

Luca Hergan and Joe Loparco

Marquette University

1313 W Wisconsin Ave, Milwaukee, WI 53233

luca.hergan@marquette.edu joseph.loparco@marquette.edu

Abstract

This paper aims to investigate the relationship between laptop price and various laptop features. We employed the use of hot encoding to convert categorical data types to numerical. We go about this by splitting our data into train/test groups and then implementing two linear regression models with said groups. The models reveal the correlations between the price and other features in our data set. We found that the main hardware components, such as RAM, CPU, and GPU have the strongest impact on price. The accuracy of our model was limited by several things such as the way we hot encoded our categorical data, the inability of linear regression to fit non-linear relationships, and the assumption of fair pricing of the laptops in our dataset across all time as components have progressed, and gotten cheaper.

1 Introduction

For our final project we choose to perform a price analysis, and then price prediction of Laptops and using a linear regression model. We aimed to use this model to determine the effect that various features have on the overall price of a laptop. There are many reasons we felt compelled to do this such as the prevalence of laptops in everyday life and especially our own lives as computer scientists. This relevance is compounded by the fact that laptops are an incredibly expensive item that have a life span of multiple years, and thus it is pertinent that we and other consumers make good choices when it is time to buy.

2 Pre-processing

2.1 Data Importing

The data set we used for the project was of relatively high quality. This meant that it did not require much pre-processing. Since our data was encoded in Windows-1252, a legacy single-byte character encoding, we were forced to use a special command for import in pandas called 'encoding'.

2.2 Data Engineering

We then proceeded to data engineer our data. The first thing that we did was check for missing elements since they could have a significant impact on our model. We did not find any missing values, reinforcing our belief in the quality of the data set. We then hot-encoded the string features to obtain numerical values (GPU/CPU/Brand/Company/Laptop type/Laptop ID/Screen resolution/OS) and turned all strings which could be represented as integers or floats into integers or floats (Memory/Weight/Price). We also added a new column that measured the total storage capacity separately from the storage models because we thought that the difference between HDD and SSD would also have an impact on price (Figure 1).

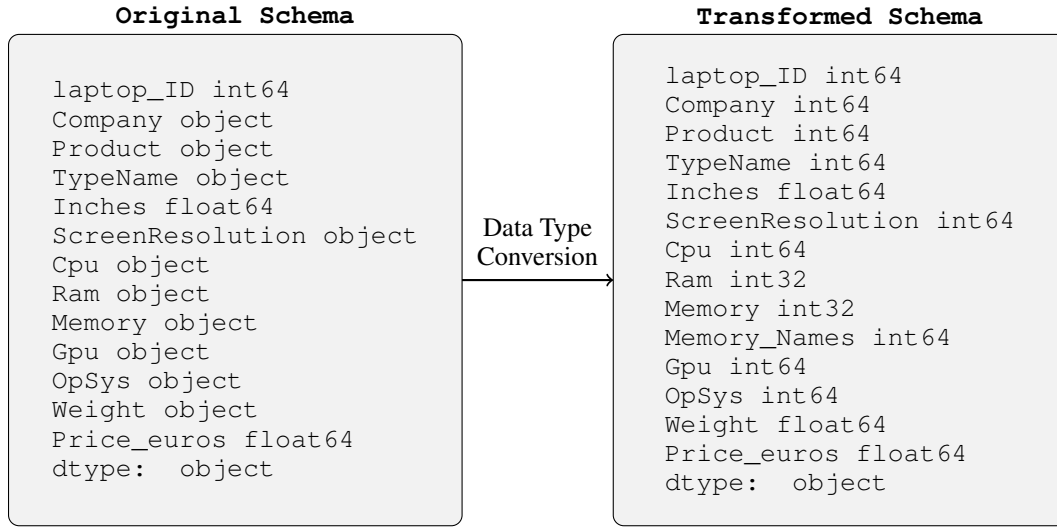


Figure 1: Data type transformation showing conversion from categorical objects to numeric encodings.

3 Our Model: Linear Regression

Using these engineered data, we then proceeded to build two different linear regression models. One used all data, including the hot encoded data, and the other only used numerical data. This was to check our suspicion that all data indeed contributed to the price, even non-numerical. (Figure 2)

3.1 All Data

The model incorporating all features, including hot-encoded categorical data, demonstrated stronger predictive performance compared to numerical data alone. With thirteen coefficients and an intercept of -181.705, this model achieved an R^2 score of 65.54%, indicating that it explains approximately two-thirds of the variance in laptop prices. The model's mean absolute error of 302.80 euros suggests reasonable accuracy in price predictions, though the root mean squared error of 429.74 euros indicates some sensitivity to outliers and larger prediction errors on higher-priced laptops.

$$\begin{aligned}
 \beta_1 &= 0.126 & \beta_8 &= 77.940 \\
 \beta_2 &= 8.405 & \beta_9 &= 4.694 \\
 \beta_3 &= 0.152 & \beta_{10} &= -0.058 \\
 \beta_4 &= 40.660 & \beta_{11} &= 3.118 \\
 \beta_5 &= -73.473 & \beta_{12} &= 103.050 \\
 \beta_6 &= 9.516 & \beta_{13} &= 136.530 \\
 \beta_7 &= 5.129 & \text{intercept} &= -181.705
 \end{aligned}$$

3.2 Numerical data

The simplified model using only numerical features showed reduced but still meaningful predictive power. With three coefficients, which included RAM and total storage (named memory), had an intercept of 389.385. This model achieved an R^2 score of 56.62%. The higher mean absolute error of 346.78 euros and root mean squared error of 482.12 euros compared to the full model suggests that categorical features like brand, GPU, and CPU specifications contain valuable information for price prediction that cannot be captured by numerical features alone.

$$\begin{aligned}
 \beta_1 &= 107.235 & \beta_3 &= -49.339 \\
 \beta_2 &= -0.118 & \text{intercept} &= 389.385
 \end{aligned}$$

All Data		Numerical Data	
Mean absolute error:	302.80	Mean absolute error:	346.78
Mean squared error:	184680.45	Mean squared error:	232443.67
Root mean squared error:	429.74	Root mean squared error:	482.12
R^2 score:	65.54%	R^2 score:	56.62%

Figure 2: Results for both linear regression models

61 4 Limitations

62 4.1 Model Limitations

63 Linear regression is a relatively simple model and as a result has drawbacks. The first significant one
64 that we encounter is its ability to measure non-linear relationships between various features. This can
65 be seen in our residual graph as the prices for higher-end laptops are very rarely predicted near their
66 true price. We believe this is due to how the pricing works in the market - a marginal gain on the
67 highest end laptops can cause a significant increase in price. Since most of the model was trained on
68 lower end laptops, it is more likely to minimize the loss for those occurrences, and so when left to
69 estimate higher-end parts it does not have appropriate weighting.

70 4.2 Data Limitations

71 Our dataset and encoding present several notable limitations that could affect model accuracy. Our
72 primary concern is our hot encoding approach for categorical variables. Here we made assumptions
73 about the ordinal nature of certain features. For instance, our encoding of operating systems {'An-
74 droid': 0, 'Chrome OS': 1, 'Linux': 2, 'Mac OS X': 3, 'No OS': 4, 'Windows 10': 5, 'Windows 10
75 S': 6, 'Windows 7': 7, 'macOS': 8}. This ordering implicitly suggests that Windows 7 is superior
76 to Windows 10, and that having no operating system is preferable to having Android or Chrome
77 OS. This is due to the increased prices by the first mentioned operating systems, and the cheaper
78 nature of the second ones listed. These encoding decisions could introduce bias into our model by
79 misrepresenting relationships between categorical variables. Additionally, our dataset might contain
80 pricing anomalies or outliers from outdated listings or incorrectly priced items, potentially skewing
81 our model's understanding of the non-linear relationship between component quality and price. This
82 is particularly problematic given that we assumed all laptops in our dataset were priced fairly relative
83 to their specifications. These data quality issues, combined with the inherent limitations of linear
84 regression in handling non-linear relationships suggest that our model might be less reliable when
85 predicting prices for high-end laptops where component quality and price relationships often become
86 especially non-linear.

87 5 Figures and Interpretations

88 5.1 Price Distribution

89 The price distribution appears to be uni-modally distributed as there is only one peak in our data. Our
90 data also skews heavily to the right due to the presence of a lower limit as nothing can have a price
91 lower than 0\$. (Figure 3)

92 5.2 Weight Distribution

93 The weight distribution is tri-modally distributed as there appear to be several clusters in our dis-
94 tribution chart where local maxima exist displayed by our curve. We are unsure as to why this
95 clustering manifested the way it did in our distribution, but one possible reason could be because
96 there are different types of laptops (i.e. Notebook, Gaming, Etc.) and these laptops are likely to
97 have mean weights similar to other of their type as their requirements align in terms of features and
98 consequentially categorization. One example of this would be gaming laptops which would require

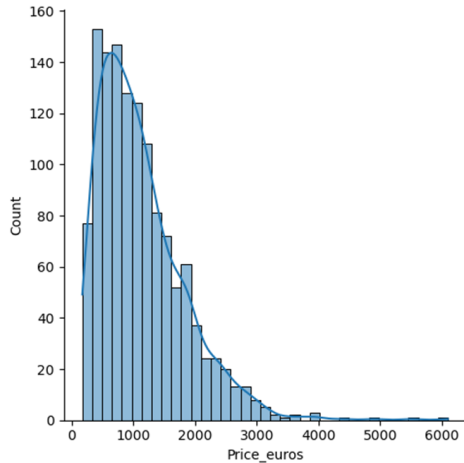


Figure 3: Price Distribution

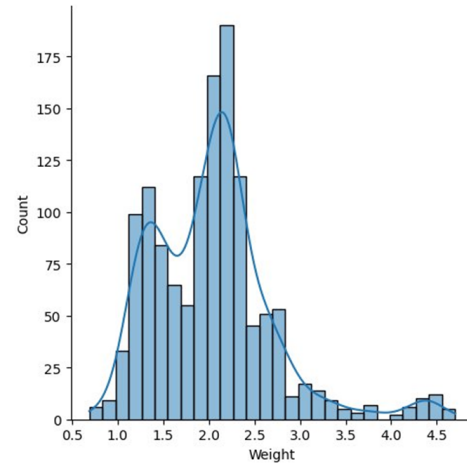


Figure 4: Weight Distribution

99 more robust hardware and therefore are more likely to define the third local maximum in weight
100 along with workstations. (Figure 4)

101 5.3 Confusion Matrix

102 Above you can see the confusion matrix which shows the linear relationship between the various
103 features of our data set. The price column is the column that pertains to our analysis. As you
104 can see in the figure above the feature that had the strongest correlation with price was RAM
105 (measured by capacity in GB not speed in mHz), followed by CPU model (categorically) and then
106 GPU (categorically). This is what we expected prior to our analysis as these are the most expensive
107 individual components in the laptop, and thus it makes sense that an increase in the quality of these
108 components would lead to a significant increase in price. RAM had a correlation coefficient of .743.
109 This shows that RAM capacity directly and linearly contributes a significant amount to the final price
110 of a laptop. (Figure 5)

111 5.4 Residual Plots

112 The distribution of our residuals appears to be approximately normal and uni-modal with a slight
113 skew right. The chart of residuals vs predicted values appears to be shaped how we would expect,
114 with greater residuals and therefore inaccuracy on higher priced laptops. We think this is due to the
115 marginal performance gain relative to increase in price for higher end products. (Figure 6)

116 6 Conclusion

117 In this project we conducted a statistical analysis on the price of laptops, and which factors contribute
118 most to said price. Our motivation for this project was the prevalence of laptops in our personal and
119 educational lives. We aimed to predict the price of laptops by training and testing two separate linear
120 regression models. We did this by splitting our pre-processed data into train and test categories, and
121 assessing our performance on the test data after fitting our models. Our results showed us that the
122 main hardware components, RAM, CPU, and GPU in decreasing order, have the strongest correlation
123 to the price of a laptop. Despite our models being relatively accurate, we ran into several limitations
124 throughout the course of this project. These limitations range from the inability of our model to fit
125 onto non-linear relationships. Adding to this, our model was also limited by our implementation of
126 hot-encoding and the assumption of fair pricing close to market suggested retail price. In conclusion
127 the price of a laptop is affected by many features but is most greatly affected by the choice of hardware
128 components.

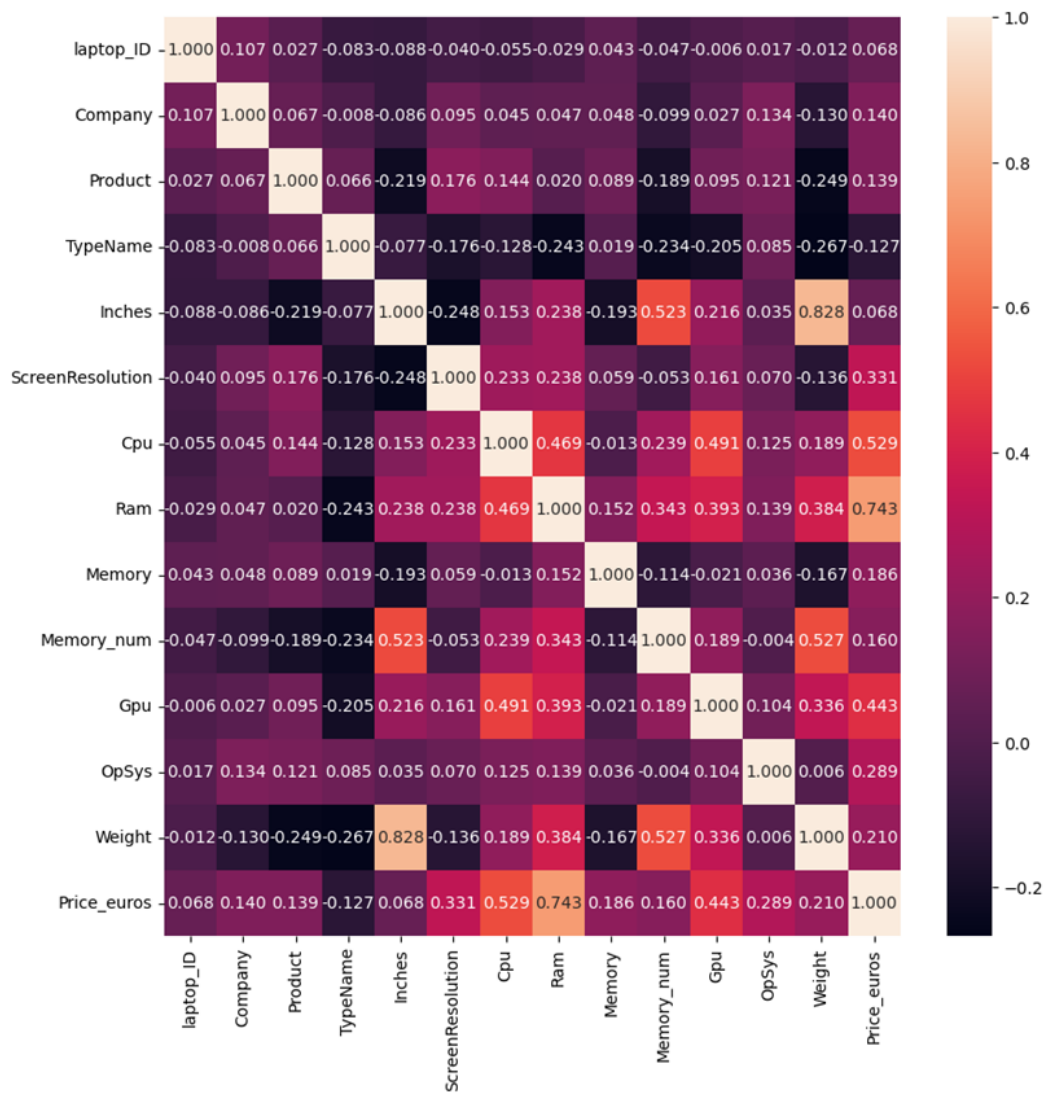


Figure 5: Confusion Matrix

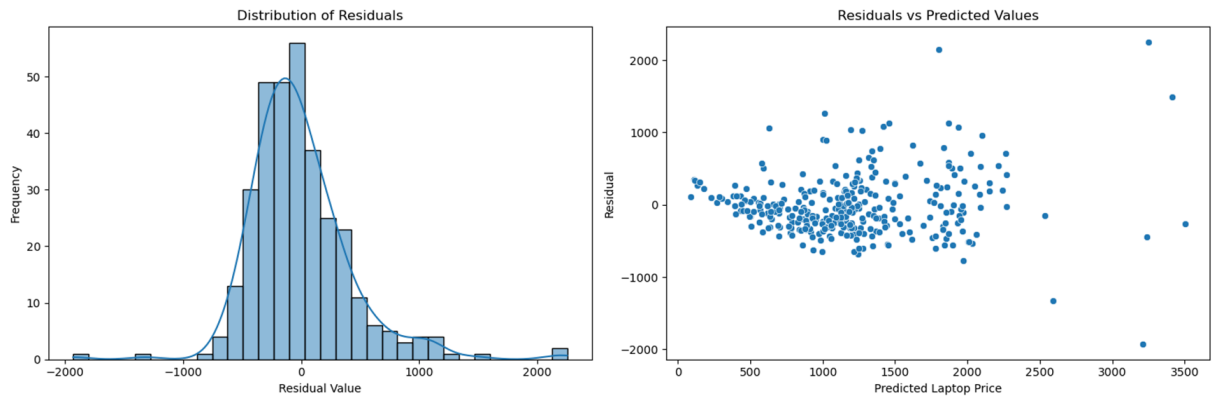


Figure 6: Plot of Residuals